NVIDIA: validation and AI use

Technical evidence | Prepared October 19, 2026 | Historical cutoff: September 10 close

15/15 finance checks pass. Published known answers, source definitions, statement articulation and failure injections target financial claims. Passing tests establish implementation consistency; they do not establish that forecasts are true or that missing human events occurred.

Evidence boundary and independent benchmarks

The latest financial date is July 26, 2026. Annual/interim periods, publication dates, units, transformations and exclusions are mapped in data/source-map.md. Facts, normalizations, forecasts and outputs are separated in data/inputs.json and outputs/results.json. No October market observations enter the valuation. Engine runtime needs Python 3.10+ and the standard library; optional PDF-authoring packages are separate.

The Week 3 published training answers independently specify operating EV **$1,624.86852 million** and **$27.49737/share**; all nine published sensitivity cells reconcile within tolerance. The Week 2 fixture independently specifies **FCFF $77.5 million**, common equity **$820 million** and **$8.20/share**. These are course benchmarks, not expected values generated by this model. Benchmark paths: lessons/week-03/teach-dcf-worked-example.md and lessons/week-02/starter/mini_case.csv.

Observed technical validation

Check / claim	Result	Observed evidence
1. Training DCF and nine cells	PASS	EV $1,624.86852m; $27.49737/share; all nine published cells
2. Training FCFF and bridge	PASS	FCFF $77.5m; equity $820m; $8.20/share
3. Historical reconciliation	PASS	Three income/cash periods; two balance dates; zero income/cash reconciliation error
4. Independent source definitions	PASS	Issuer release vs SEC statements/Notes 5, 10, 11: cash/debt securities $56,586m; equity investments $42,783m separate; book/fair debt $33,366m/$31,400m; Treasury nominal 10-year 4.95%
5. Forecast articulation	PASS	25 annual forecasts checked across cases/what-ifs; maximum error $2.33e-10m; cash from cash flow and equity from retained earnings, no balancing plug
6. Prewritten changed input	PASS	WACC +1pp: $118.58 to $108.38; expected and actual direction lower; WATCH / DEFER persists; prediction AI-authored
7. WACC/g monotonicity	PASS	Higher WACC lowers value; higher g raises value after terminal reinvestment and full bridge recomputation
8. Terminal/unit boundaries	PASS	Reject g >= WACC, rates entered as whole percentages, and g >= terminal ROIC

Failure tests and decision sensitivity

Check / claim	Result	Observed evidence
9. EV/equity/share bridge	PASS	Common equity $2,874.73bn; 2% more shares lowers per-share value; dividend cash recomputed
10. Source/liquidity corruption	PASS	Deliberate $1m asset corruption fails; forced cash deficit fails with explicit funding message
11. Investment contamination	PASS	Exclude $103,021m financial portfolio from operating NWC. Wrong scaling creates $126,463.73m discounted false investment; bounded $5.22/share drag, not a complete alternative valuation
12. Signed cash flows	PASS	Negative first-year FCFF lowers EV by $90.9091m; no removal of negative flows
13. SBC owner-cost challenge	PASS	Wrong addback inflates five-year value by at least $1.05/share; terminal error would add more. Correct model retains cost through cash replacement
14. Peer consistency	PASS	Lab 8 policy precedes selection; annual GAAP EPS/revenue; consistent cash/debt/share definitions and operating-lease treatment
15. Downside / reverse DCF	PASS	Low case changes WATCH / DEFER to DO NOT INITIATE. Conditional market-implied solution adds 19.82pp to each Data Center growth rate

Complete machine-readable observations, tolerances and fixtures: outputs/validation-results.json and tests.py. The engine exits nonzero on failed checks. Important caught messages include **"BS_Jul 26 assets: 320273 != 320272"** and **"base year 1 cash below floor; explicit funding needed"**. A passing failure test means the deliberately bad input was detected, not that an error was ignored.

Clean-environment cold run and first failure

The actual assistant cold run copied the project and created a new virtual environment with **--without-pip**. It used no inherited third-party packages, network access or credentials. Four deterministic outputs matched byte-for-byte: results.json, validation-results.json, visible-output.md and dashboard.html. The receipt records Python 3.13.15, actual timing and hashes. First failure: **none observed**; no repair event was invented. Evidence: evidence/cold-run-log.md and evidence/cold-run-receipt.json.

Video 2 describes a student run, but full audiovisual execution/ownership has not been independently reviewed. The assistant receipt does not prove a student performance. README commands reproduce the visible result; changed runs require a separate output directory to protect frozen results.

Why limitations change the action

Terminal value is 65.1% of EV. WACC +1pp reduces base value to $108.38, still below $218.36, so the action stays WATCH / DEFER. The combined demand/financing downside changes the action to DO NOT INITIATE. These are different tests: an unchanged threshold decision is explained, while the risk regime can reverse the recommendation.

Prewritten directional test versus required human lock

The AI-authored prediction was committed **before** the rebuilt-model run at commit **7c404d5**; see evidence/directional-test-precommit.md. WACC changes from **0.138965097** to **0.148965097**, decimal units (+1 percentage point). Expected lower EV/share value and continued deferral match the observed $118.58 to $108.38 and unchanged WATCH / DEFER. This is a valid prewritten technical test, **not a completed human-authored AI-off lock**.

Locked Changed-Input Record field	Current evidence
Human prediction / exact input	Student original text d9f1f52; 3.0% to 2.5% yearly (-50bp), with lower value and unchanged WATCH / DEFER predicted
Immutable specification and time	5a9de1d, October 19 at 20:54:56 UTC; run started 20:55:33 UTC
Baseline / changed result	$118.58487 / $116.56710 per share; -$2.01777 (-1.70%); both WATCH / DEFER
Decision effect / comparison	Market $218.36 remains above changed entry ceiling $93.25; direction/action match; "much lower" was not a quantified magnitude target
Mechanism / ownership	Five-year forecast fixed. Reinvestment falls 12% to 10%, offsetting part of the lower-growth valuation effect. Student prediction; assistant execution

Status: **EXECUTED**. Full outputs, hashes, commands and prediction chronology: locked record and run receipt. The 1.70% decrease is modest. Instructor permission for AI-assisted wording is student-reported; no independent AI-off attestation is invented. The earlier AI WACC test remains separate.

Contamination catalog and financial limits

- **Base-year normalization:** FY26 net H20 charge $4,320m; normalized EBIT $134,707m and gross margin 73.07%. It has already rolled out of the LTM base, so it is not added again.
- **Operating boundary:** Investment gains and interest income are excluded from operating FCFF; equity holdings are not cash or working capital. All signed operating cash flows remain in the DCF.
- **Share/price basis:** Annual weighted diluted shares are not exact current shares; the 24,242m diluted proxy is explicit. Old intraday $224.35 differs from September 10 close $218.36.
- **Owner cost/acquisitions:** SBC and retention retain their cost. Pending acquisition cash is included without operating upside; provisional goodwill is not a final GAAP allocation.
- **Forecast limits:** Cash interpolation, leases, deferred tax and refinancing are simplified; terminal returns and recoveries are assumptions. Guarantee bridge losses are not booked liabilities; actual calls require funded forecasts.

Assumption challenges and partner review

Credit: **peer challenge**, October 1, 13:40-14:06 EDT; student-confirmed Antigravity log. Demand evidence is qualified; range-bias critique accepted. Independent equal-relative tests give margin $14.04/share span vs DC growth $8.89. See attributed peer review and evidence/assumption-challenges.md. Independent sources/tests decide which outputs survive.

Material AI uses and dispositions

Interface: **Codex based on GPT-6**; exact backend build/version is not exposed. Interaction date: **October 19, 2026**. No model identity or hidden history is invented for older labs.

AI use	Independent evidence / disposition	Decision consequence
Inspect Edition A and Labs 5/8/10/11/12	File hashes and prior commits; ACCEPTED WITH QUALIFICATION; original checkpoint receipt missing	Research admission distinguished from buying
Extract SEC, issuer and Treasury facts	Statement/notes and Treasury-column reconciliation; CORRECTED old investment/debt classification	Avoid false reinvestment and inflated cash bridge
Build statements, FCFF, WACC, bridge and UI	Published answers, articulation, sign/boundary/failure checks; ACCEPTED after 15 passing tests	Conditional valuation range and usable output
Draft forecasts, stresses and peer narrative	Driver challenges, guidance boundary and peer definitions; QUALIFIED	Downside gate, watch/defer and no mechanical averaging
Antigravity discussion record (use reported October 1; model/version not supplied)	Student confirms actual peer discussion; independent Lab 11 rerun and equal-shock follow-up; ACCEPTED WITH CORRECTIONS	Attributed challenge; range-dependent ranking and demand conditions
Draft documents, scripts and editorial transcript corrections; run QA/freeze	Numerical comparison, preserved edit log, PDF rendering and public hash checks; ACCEPTED as assistance	Clearer communication; no certified audio or human authorship

Recording and transcript status

Video / purpose	Supplied link	Duration
1. Research evolution	Open Video 1	2:53 / max 3:00
2. Code and validation	Open Video 2	2:58 / max 5:00
3. Committee decision	Open Video 3	3:00 / max 5:00

Anonymous metadata confirms player status OK and durations; full playback is unreviewed. Student-requested transcript corrections and hashes are logged in evidence/video-review.json. Audio alignment/timestamp indices remain unverified; original downloads and prior snapshots are preserved.

Freeze, access and remaining submission evidence

File-integrity.json hashes stable artifacts; the manifest has 11 required entries plus optional deployment. Publication-audit.json records public access/hash checks. Tags preserve prior versions; PDF copies match. Full sources/results are visible without execution.

Still unresolved: original checkpoint PDF/receipt, audio/timestamp checks. Student-predicted growth test and attributed peer review are supplied. Save the final Brightspace receipt after submission; GitHub alone is not submission. Due **October 19, 2026, 11:59 PM course time**. Technical PASS is not a certified score.